

Title

Neurophysiological and Psychometric Characterization of Modality-Specific Contributions to Motor Interface Learning

Abstract

A secondary analysis of the comparative interface study examined individualized neurophysiological patterns and learning trajectories across signal modalities. Electroencephalographic signal variance and near-infrared spectroscopy hemodynamic response were analyzed using genetic algorithm-based feature selection. Fronto-central regions (electrode positions C3 and C4) consistently emerged as highest-information sites across participants (93% inclusion frequency). Near-infrared sensor arrays required heterogeneous positioning strategies, with optimal channel locations varying significantly between subjects (inter-individual distance up to 4 cm). Subgroup analysis by age, baseline attention capacity, and prior neuroimaging exposure revealed no statistically significant moderating effects on either modality's utility (all $p > 0.08$). Learning curve analysis showed equivalent trajectory steepness between groups (slopes: 0.018 versus 0.017 accuracy units per session). Across economic feasibility criteria, single-modality systems demonstrated reduced setup time and equipment cost without performance compromise.